

National University of Kyiv-Mohyla Academy

Faculty of Informatics

Homework #2

Dataset Collection

Brainrot analysis

Course: Computational Social Science
Team: FightClub
Lecturer: Andrew Kurochkin

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1 Papers Research

Before settling on our research theme and collecting data for it, we looked up other papers related to our general research theme to analyze how they structured their research, what data sources they used, and what they concluded. This way, we avoid researching something that has already been researched.

To find these papers we used Google Scholar, arXiv, Connected Papers, and looked through cited papers. In total we analyzed about 20 papers related to the “Brainrot” and algorithmic “rabbit holes” themes. This review gave us the following.

Most papers followed the same structure. They tell the reader what the paper is about (Abstract and Introduction), explain the research question and how they approached it (Methods), present the results, and discuss what those results mean (Discussion and Conclusion).

Abstract

Introduction

Methods

Result

Discussion

Conclusion

Data Availability

References

Funding

Figure 1: Typical section structure of the reviewed papers.

The “Methods” sections were the most useful for us. They showed that, apart from a couple of exceptions, there is no actual research on “Brainrot” based on large amounts of data. Most papers either relied on small questionnaires and surveys, or based their conclusions on existing research. The main exception we found was *“Operationalizing brain rot on TikTok through linguistic, structural, and behavioral indicators of low-cognitive-value media content”* (2026), where 78,452 TikToks were analyzed to try to define “Brainrot” through the *“identification of key textual, structural, and behavioral markers associated with low-cognitive-value media content”*.

Papers mostly agree that “Brainrot” is linked to poorer attention and wellbeing. However, this evidence comes mostly from self-reported surveys taken at a single point in time, so it cannot show whether the harm builds up over time. We found no study that tracks whether people’s own writing becomes shorter and simpler over time, or whether crises increase doomscrolling hours. This is the gap our project targets.

2 Research Topic

2.1 Main question

The impact of doomscrolling and brain rot on people, their wellbeing and cognitive abilities over time.

We want to understand the algorithms and how brain rot affects us, in order to protect our cognitive abilities.

2.2 Hypotheses and criteria

- **H1:** Discourse is getting shorter and simpler
 - Lexical & NLP approach
 - Temporal/Behavioral aspect
- **H2:** Short-form platforms accelerated the degradation of discourse
- **H3:** Wellbeing is getting worse among doomscrollers
- **H4:** Crises intensify doomscrolling and negativity

3 Dataset selection

For H1 we use a Reddit dataset spanning from 01.2012 to 06.2025, in order to also capture the period before short-form content.

Subreddit selection:

1. Short-form: r/memes, r/teenagers
2. Long-form: r/books, r/explainlikeimfive
3. Baseline: r/todayilearned
4. Qualitative: r/nosurf

For H2, H3 and H4 we use statistical data from the following Kaggle datasets:

- Reels/Shorts Consumption vs Attention Span
- Social Media Addiction dataset
- The Dark At The End Of The Tunnel: Doomscrolling On Social Media Newsfeeds
- Sleep & Doomscrolling Habits Dataset
- Social Media and Mental Health
- Student Social Media & Mental Health

Sleep & Doomscrolling Habits Dataset

"1,000 synthetic records linking doomscrolling habits to sleep quality"



[Data Card](#) [Code \(16\)](#) [Discussion \(0\)](#) [Suggestions \(0\)](#)

About Dataset

This dataset simulates 1,000 individuals answering questions about their nighttime device use, doomscrolling behavior, sleep patterns, and related psychological indicators (anxiety, stress, fatigue). It explores a topic with strong current relevance: how late-night screen habits and negative-news consumption relate to sleep quality — an area of growing public health and wellness interest.

The dataset uses a numeric-first design: 18 of 30 columns (60%) are integer or float features, ideal for correlation analysis, scaling/standardization, and numeric feature engineering, alongside 12 categorical columns for encoding practice.

Usability ⓘ

9.41

License

[CC0: Public Domain](#)

Expected update frequency

Annually

Tags

Figure 2: Kaggle data card of the Sleep & Doomscrolling Habits Dataset.

4 Downloading & Pre-processing

4.1 Reddit

Since we want to cover 13 years of Reddit history, we used the course hard drive as the source of the data.

To download it, we wrote a Python script which read each month directly from the drive and decompressed it in RAM in small chunks of 64MB, without unpacking it to disk. In RAM it kept only records from our six subreddits and only the fields needed for analysis (IDs, author, timestamp, score and text). The results were saved as compressed Parquet files, one per subreddit per month, reducing about 1.4TB of raw dumps to about 7.6 GB.

We optimized the pre-processing algorithm in three iterations and reached a processing speed of 3–5 minutes per month. We managed to download the data from 2012 to 2016, but then we hit a month which took 6 hours to process. As our investigation showed, the hard drive has damaged areas that may take hours to read or cannot be read at all. When the hard drive reaches a damaged area, it tries to read it many times in a row; this may succeed, but only after hours, or the drive may get stuck there forever. We analyzed the hard drive in CrystalDiskInfo and found that there are 374 damaged sectors on the disk and that the disk status is “Bad”. This is the worst possible status in CrystalDiskInfo.

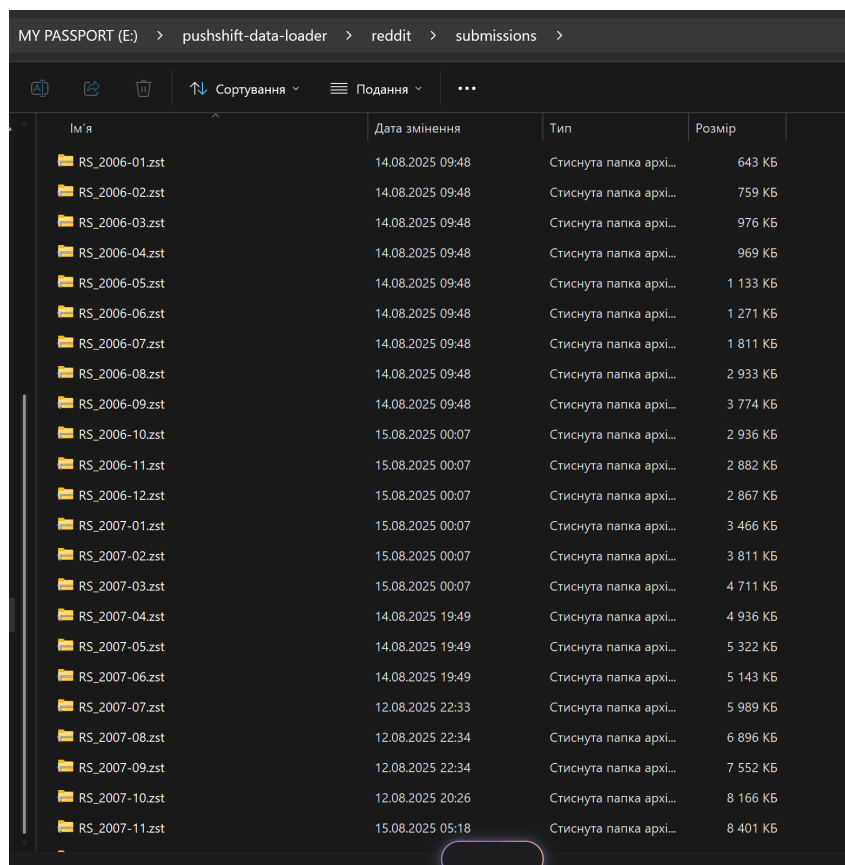


Figure 3: Compressed monthly Reddit dumps on the hard drive.

Ім'я	Стан	10% ЦП	86% Пам'ять	24% Диск	0% Мережа
Python (5)		3,8%	5 307,2 МБ	43,9 Мбіт...	0 Мбіт/с
Python		0,8%	2 339,3 МБ	13,8 Мбіт...	0 Мбіт/с
Python		1,8%	2 277,5 МБ	13,6 Мбіт...	0 Мбіт/с
Python		1,3%	682,3 МБ	10,4 Мбіт...	0 Мбіт/с
Python		0%	7,6 МБ	6,2 Мбіт/с	0 Мбіт/с
Python		0%	0,5 МБ	0 Мбіт/с	0 Мбіт/с

Figure 4: Python multiprocessing: parallel worker processes in Task Manager.



Figure 5: Frozen hard drive: 100% active time with zero read speed.

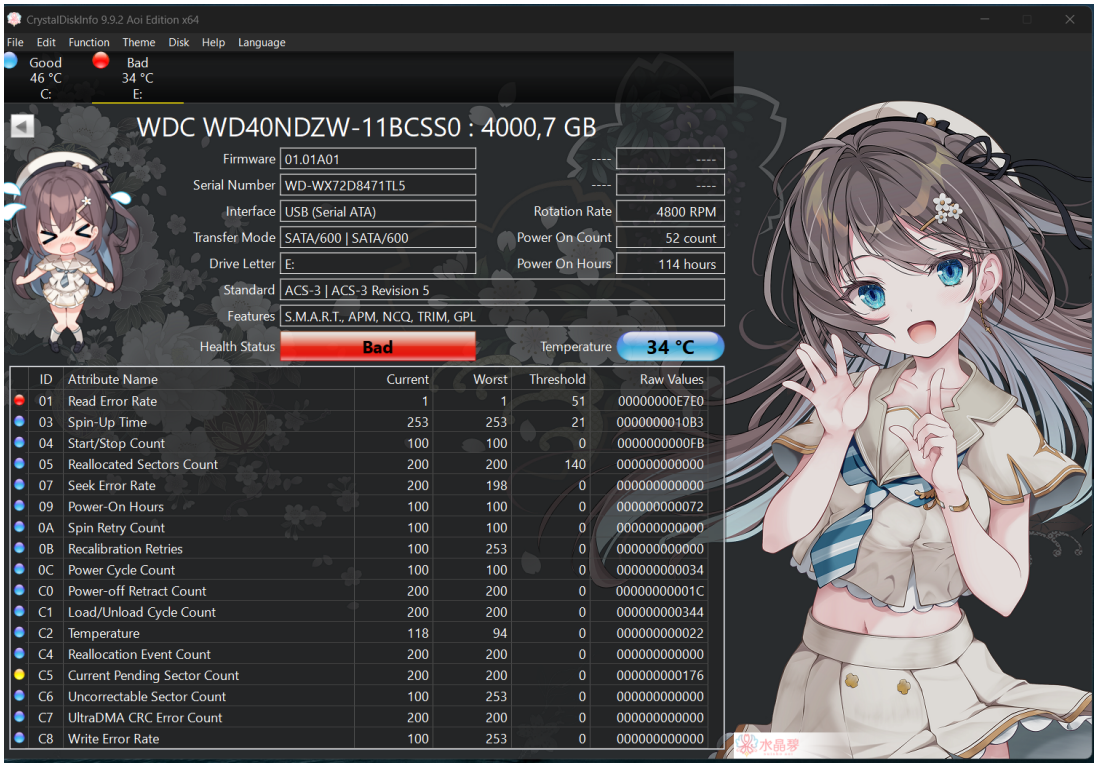


Figure 6: “Bad” health status of the hard drive in CrystalDiskInfo (374 pending sectors).

```

[02:10:11] RS_2014-08.zst      58s books=4,649, explainlikeimfive=17,891, memes=1,307, nosurf=10, teenagers=8,220, todayilearned=17,542
progress 56.2 GB / 3.1 TB ETA 11.2 h
[02:10:42] RC_2014-06.zst      162s books=66,135, explainlikeimfive=176,826, memes=1,926, nosurf=41, teenagers=274,730, todayilearned=447,318
progress 59.9 GB / 3.1 TB ETA 11.0 h
[02:13:21] RS_2014-09.zst      176s no rows <-- ERROR: PermissionError: [Errno 13] Permission denied
progress 60.8 GB / 3.1 TB ETA 13.0 h
[02:15:04] RC_2014-07.zst      343s books=80,832, explainlikeimfive=208,488, memes=1,777, nosurf=43, teenagers=353,456, todayilearned=509,271
progress 65.0 GB / 3.1 TB ETA 13.5 h
[02:15:28] RC_2014-08.zst      302s books=71,038, explainlikeimfive=190,492, memes=1,893, nosurf=48, teenagers=128,032, todayilearned=505,328
progress 69.2 GB / 3.1 TB ETA 13.0 h
[02:15:30] RS_2014-10.zst      28s books=4,140, explainlikeimfive=17,950, memes=1,179, nosurf=6, teenagers=7,246, todayilearned=16,541
progress 70.1 GB / 3.1 TB ETA 12.8 h
[02:15:45] RC_2014-09.zst      127s books=68,427, explainlikeimfive=178,618, memes=1,658, nosurf=55, teenagers=88,418, todayilearned=444,844
progress 74.2 GB / 3.1 TB ETA 12.3 h
[02:15:46] RS_2014-11.zst      28s books=4,526, explainlikeimfive=19,213, memes=1,266, nosurf=8, teenagers=6,631, todayilearned=15,127
progress 75.1 GB / 3.1 TB ETA 12.1 h
[02:16:12] RS_2014-12.zst      30s books=5,288, explainlikeimfive=20,175, memes=1,354, nosurf=9, teenagers=7,499, todayilearned=16,225
progress 76.0 GB / 3.1 TB ETA 12.3 h
[02:17:41] RC_2014-10.zst      134s books=55,050, explainlikeimfive=189,797, memes=1,979, nosurf=19, teenagers=88,476, todayilearned=467,272
progress 80.3 GB / 3.1 TB ETA 12.5 h
[02:17:57] RC_2014-11.zst      134s books=63,871, explainlikeimfive=182,657, memes=1,894, nosurf=27, teenagers=96,400, todayilearned=426,986
progress 84.4 GB / 3.1 TB ETA 12.1 h
[02:18:04] RS_2015-01.zst      35s books=5,749, explainlikeimfive=21,666, memes=1,329, nosurf=7, teenagers=8,219, todayilearned=20,187
progress 85.6 GB / 3.1 TB ETA 12.0 h
[02:18:40] RC_2014-12.zst      140s books=71,930, explainlikeimfive=218,766, memes=2,072, nosurf=38, teenagers=124,408, todayilearned=489,050
progress 90.0 GB / 3.1 TB ETA 11.8 h
[02:18:50] RS_2015-02.zst      34s books=4,559, explainlikeimfive=19,506, memes=1,352, nosurf=11, teenagers=5,493, todayilearned=18,175
progress 91.0 GB / 3.1 TB ETA 11.7 h
[02:19:21] RS_2015-03.zst      43s books=4,919, explainlikeimfive=20,734, memes=1,298, nosurf=26, teenagers=5,949, todayilearned=19,140
progress 92.4 GB / 3.1 TB ETA 11.8 h
[02:20:45] RC_2015-01.zst      163s books=78,576, explainlikeimfive=233,888, memes=2,339, nosurf=59, teenagers=154,319, todayilearned=599,295
progress 97.3 GB / 3.1 TB ETA 11.9 h
[02:21:40] RS_2015-04.zst      74s no rows <-- ERROR: PermissionError: [Errno 13] Permission denied
progress 98.8 GB / 3.1 TB ETA 12.2 h
[02:22:25] RC_2015-02.zst      217s books=57,680, explainlikeimfive=199,512, memes=1,952, nosurf=34, teenagers=74,685, todayilearned=489,841
progress 103.3 GB / 3.1 TB ETA 12.0 h
[02:30:14] RS_2015-05.zst      486s no rows <-- ERROR: PermissionError: [Errno 13] Permission denied

```

Figure 7: Reddit processing logs, including months that failed with read errors.

4.2 Kaggle

To prepare these datasets, we wrote a cleaning script. For each file it removes duplicate rows, reports missing values, normalizes text and date fields, and adds a column showing which dataset each row came from. In the end, only 3 duplicates and 1 empty row had to be removed across the three full datasets.

```

--- cleaning Doomscrolling_Study2_Dataset.csv ---
[Doomscrolling_Study2_Dataset.csv] exact duplicate rows removed: 0
[Doomscrolling_Study2_Dataset.csv] duplicate 'ResponseId' rows removed: 0
[Doomscrolling_Study2_Dataset.csv] wrote 401 rows -> data/clean/Doomscroll

--- cleaning Doomscrolling_Study3_Dataset.csv ---
[Doomscrolling_Study3_Dataset.csv] exact duplicate rows removed: 0
[Doomscrolling_Study3_Dataset.csv] duplicate 'ResponseId' rows removed: 0
[Doomscrolling_Study3_Dataset.csv] wrote 502 rows -> data/clean/Doomscroll

--- cleaning Social_Media_Addiction.csv ---
[Social_Media_Addiction.csv] dropped 1 fully-null row(s) (no usable data)
[Social_Media_Addiction.csv] exact duplicate rows removed: 3
[Social_Media_Addiction.csv] duplicate rows by content-hash removed: 0
[Social_Media_Addiction.csv] 8 row(s) had null gender -> labeled 'Unknown'
)
[Social_Media_Addiction.csv] merged 35 'Insta' row(s) into 'Instagram' (pe
[Social_Media_Addiction.csv] wrote 1199 rows -> data/clean/Social_Media_Ad

--- cleaning reels_attention_span_dataset_12000.csv ---
[reels_attention_span_dataset_12000.csv] exact duplicate rows removed: 0
[reels_attention_span_dataset_12000.csv] duplicate 'user_id' rows removed:
[reels_attention_span_dataset_12000.csv] wrote 12000 rows -> data/clean/re
t_12000.csv

```

Figure 8: Console output of the cleaning script for the Doomscrolling, Social Media Addiction and Reels datasets.

How it works: since each source has its own schema, the script runs a different cleaning function per file. For every file it first removes duplicate rows, then removes duplicates by whatever unique identifier exists (`ResponseId`, `user_id`) or, for `Social_Media_Addiction`, by a hash of the row's content, since it has no ID. Text fields are stripped of extra whitespace and Unicode-normalized (NFKC) throughout. The two Doomscrolling survey files additionally get their date columns parsed to proper datetimes. `Social_Media_Addiction` gets extra fixes:

one fully empty row is dropped before deduplication, missing gender values are changed to “Unknown”, “Insta” is changed to “Instagram”, and age is converted to integer. In the end, every action is logged to the console.

We cleaned the other three datasets by first removing duplicate rows and keeping only the features that are relevant to the H3 hypothesis about social media use, doomscrolling, and wellbeing. We kept variables related to screen and social media use, doomscrolling, sleep, stress, anxiety, mental health, and a few basic personal factors such as age and gender. We converted numerical columns to the correct format and checked for impossible values, such as negative screen time, sleep time above 24 hours, or unrealistic ages. We also removed extra spaces from text values and changed invalid numerical values to missing values instead of deleting the whole row. We did not combine the three datasets because they contain different types of information and participants, so each dataset was cleaned separately according to its purpose in the research.

Результати

```
ЗВІТ ОЧИЩЕННЯ: sleep_doomscrolling_habits
Розмір до: (1000, 30)
Розмір після: (1000, 30)
- Дублікати за respondent_id: 0
- Повні дублікати: 0
- Doomscrolling за ніч >12 год: 0
- Сон + екранний час > 30 год: 0
- caffeine_intake_mg_per_day: заповнено пропусків: 50
- sleep_quality_score: заповнено пропусків: 40
- exercise_minutes_per_day: заповнено пропусків: 50
- occupation_status: заповнено пропусків: 30
- primary_device_used_at_night: заповнено пропусків: 30
- bedtime_routine_type: заповнено пропусків: 20
- bedtime_screen_time_minutes: IQR-викиди (обрізано): 3
- total_daily_screen_time_hours: IQR-викиди (обрізано): 1
- doomscroll_sessions_per_night: IQR-викиди (обрізано): 1
- avg_doomscroll_session_minutes: IQR-викиди (обрізано): 4
- number_of_night_wakeups: IQR-викиди (обрізано): 2
- caffeine_intake_mg_per_day: IQR-викиди (обрізано): 2
- days_since_last_digital_detox: IQR-викиди (обрізано): 8

ЗВІТ ОЧИЩЕННЯ: smmh
Розмір до: (481, 21)
Розмір після: (481, 31)
- Повні дублікати: 0
- One-hot колонок для платформ: 9
- organization_type: заповнено пропусків: 30

ЗВІТ ОЧИЩЕННЯ: Student Social Media And Mental Health Impact
Розмір до: (5000, 13)
Розмір після: (4998, 14)
- УВАГА stress_level: не числова колонка, конвертацію пропущено:
5000/5000 значень не є числами; приклади: ['Medium', 'Low', 'Very
High', 'High']
- stress_level: унікальні значення: ['High', 'Low', 'Medium',
'Very High']
- Повні дублікати: 2
- physical_activity_hours: поза діапазоном [0, 12]: 10
- Сума годин на добу > 30: 0
- physical_activity_hours: заповнено пропусків: 10
- stress_level_num: заповнено пропусків: 1621
```

Figure 9: Cleaning report for the Sleep & Doomscrolling Habits, Social Media and Mental Health, and Student Social Media & Mental Health datasets.

5 Exploration

5.1 Data Frames

5.1.1 Kaggle

1. Reels/Shorts Consumption vs Attention Span

reels_attention_span_dataset_12000: (12000, 11)

	user_id	age	reels_watch_time_hours	daily_screen_time_hours	sleep_hours	attention_span_score	focus_level	task_completion_rate	stress_level	platform	source_dataset
0	1	21	1.86	7.34	8.95	4.99	5.09	45.44	Medium	TikTok	reels_attention_span_dataset_12000.csv
1	2	34	3.03	5.78	5.78	3.73	8.25	50.17	High	TikTok	reels_attention_span_dataset_12000.csv
2	3	43	3.96	9.62	5.46	5.48	8.96	43.27	Low	TikTok	reels_attention_span_dataset_12000.csv
3	4	29	3.56	7.25	5.09	5.54	9.77	64.99	Low	Instagram Reels	reels_attention_span_dataset_12000.csv
4	5	25	4.58	2.13	8.12	8.07	4.11	89.28	Low	Instagram Reels	reels_attention_span_dataset_12000.csv

Figure 10: First rows of the Reels/Shorts Consumption vs Attention Span dataset (12,000 rows, 11 columns).

2. Social Media Addiction dataset

Social_Media_Addiction: (1199, 11)

	age	gender	daily_social_media_hours	platform_usage	sleep_hours	screen_time_before_sleep	CGPA	physical_activity	social_interaction_level	addiction_level	source_dataset
0	14	male	7.9	Instagram	7.4	2.9	3.01	1.5	low	7.6	Social_Media_Addiction.csv
1	19	female	1.9	TikTok	8.0	2.9	3.22	0.8	high	2.8	Social_Media_Addiction.csv
2	17	female	1.3	Instagram	7.6	0.5	3.92	0.0	high	1.9	Social_Media_Addiction.csv
3	15	male	7.4	TikTok	6.9	1.6	3.48	0.8	medium	6.5	Social_Media_Addiction.csv
4	15	female	4.7	Both	4.9	3.0	2.37	1.4	medium	6.6	Social_Media_Addiction.csv

Figure 11: First rows of the Social Media Addiction dataset (1,199 rows, 11 columns).

3. The Dark At The End Of The Tunnel: Doomscrolling On Social Media Newsfeeds

Doomscrolling_Study2_Dataset: (401, 93)

	StartDate	EndDate	Progress	Duration_in_seconds	RecordedDate	ResponseId	Consent	DS1.1	DS1.2	DS1.3	...	DS8.7	DS8.8	DS8.9	DS8.10	Gender	Gender_3_TEXT	Age	Nation	Edu	source_dataset
0	2021-03-16 05:29:53	2021-03-16 05:43:12	100	799	2021-03-16 05:43:13	R_XUu0DGA1S97cnQd	1	3	2	5	...	3	5	4	3.0	2	NaN	34	45	25	Doomscrolling_Study2_Dataset.csv
1	2021-03-16 05:29:45	2021-03-16 05:38:49	100	543	2021-03-16 05:38:49	R_1eVnKQ2VAQyMa	1	7	2	2	...	7	6	7	7.0	1	NaN	28	161	16	Doomscrolling_Study2_Dataset.csv
2	2021-03-16 05:29:45	2021-03-16 05:39:43	100	597	2021-03-16 05:39:43	R_3C8SgJ1NagKm8FI	1	2	2	5	...	3	4	6	5.0	1	NaN	34	138	12	Doomscrolling_Study2_Dataset.csv
3	2021-03-16 05:29:52	2021-03-16 05:42:43	100	770	2021-03-16 05:42:43	R_279gka0ywn0VSS8	1	2	2	3	...	4	2	6	5.0	1	NaN	20	158	13	Doomscrolling_Study2_Dataset.csv
4	2021-03-16 05:29:52	2021-03-16 05:38:13	100	500	2021-03-16 05:38:13	R_0NubTwgc2K5Ag9	1	2	5	3	...	5	4	2	3.0	1	NaN	21	138	15	Doomscrolling_Study2_Dataset.csv

Figure 12: First rows of the Doomscrolling Study 2 dataset (401 rows, 93 columns).

Doomscrolling_Study3_Dataset: (502, 205)

	StartDate	EndDate	Progress	Duration_in_seconds	RecordedDate	ResponseId	Consent	BFI_2_XS_1	BFI_2_XS_2	BFI_2_XS_3	...	PolSelf_2	PolSelf_3	PolSelf_4	PolSelf_5	Gender	Gender_3_TEXT	Age	Nation	Edu	source_dataset
0	2021-03-31 06:27:47	2021-03-31 08:17:45	100	6597	2021-03-31 08:17:45	R_27HbzH6DGhqqPZ	1	5	2	5	...	1	3	1	4	1	NaN	19	137	14	Doomscrolling_Study3_Dataset.csv
1	2021-03-31 06:35:32	2021-03-31 07:42:10	100	3997	2021-03-31 07:42:11	R_2VwEp7dabNmYmN	1	3	4	4	...	7	5	5	5	2	NaN	20	137	16	Doomscrolling_Study3_Dataset.csv
2	2021-03-31 06:28:33	2021-03-31 07:31:59	100	3806	2021-03-31 07:32:00	R_PMzbtvqCDUFRyV	1	4	4	5	...	6	5	3	5	1	NaN	30	137	14	Doomscrolling_Study3_Dataset.csv
3	2021-03-31 06:32:44	2021-03-31 07:35:40	100	3776	2021-03-31 07:35:41	R_2qzwhh2GUA9go2pK	1	2	5	5	...	7	7	1	4	1	NaN	19	137	13	Doomscrolling_Study3_Dataset.csv
4	2021-03-31 06:32:45	2021-03-31 07:33:36	100	3650	2021-03-31 07:33:36	R_54GBIDMKWQcCt	1	5	5	3	...	1	1	1	2	1	NaN	20	137	14	Doomscrolling_Study3_Dataset.csv

Figure 13: First rows of the Doomscrolling Study 3 dataset (502 rows, 205 columns).

4. Sleep & Doomscrolling Habits Dataset

Sleep_doomscrolling_habits_dataset: (1000, 30)																						
	respondent_id	age	bedtime_start_time_minutes	total_daily_screen_time_hours	doomscroll_sessions_per_night	avg_doomscroll_session_duration	sleep_hours_per_night	sleep_latency_minutes	number_of_night_wakings	caffeine_intake_avg_per_day	...	occupation_status	country_region	primary_device_used_at_night	bedtime_routine_type	sleep_night_awake	keeps_phone_in_bedroom	experiences_negative_emotions_casualty	uses_doomscrolling_app	doomscrolls	sleep_quality_category	feedback
0	00000	23.0	110	1.65	0.0	0.0	6.1	18.0	0.0	10.0	...	Employed	Germany	Smartphone	Meditation/Listening	False	True	True	False	True	Good	
1	00001	35.0	90.0	1.01	1.0	10.0	7.0	10.0	0.0	10.0	...	Student	India	Smartphone	Meditation/Listening	False	True	True	False	True	Good	
2	00002	27.0	110	1.15	1.0	10.0	7.0	10.0	0.0	10.0	...	Student	India	Smartphone	Reading	False	True	False	True	True	Good	
3	00003	27.0	110	1.14	0.0	10.0	6.4	10.0	0.0	10.0	...	Employed-Pub	United States	Smartphone	No Fixed Routine	False	True	True	False	False	Fair	
4	00004	17.0	90.0	4.00	1.0	10.0	7.2	18.0	1.0	76.0	...	Employed Part-Time	United States	Smartphone	Watching Videos	False	False	False	False	False	Fair	

Figure 14: First rows of the Sleep & Doomscrolling Habits dataset (1,000 rows, 30 columns).

5. Social Media and Mental Health

sm_clean: (481, 31)

	timestamp	age	gender	relationship_status	occupation_status	organization_type	uses_social_media	platforms	avg_daily_time_social_media	use_without_purpose	platform_discord	platform_facebook	platform_instagram	platform_pinterest	platform reddit	platform_snapchat	platform_ tiktok	platform_twitter	platform_youtube	avg_daily_hours_sm
0	2022-04-18 18:18:47	21.0	Male	In A Relationship	University Student	University	True	Facebook, Twitter, Instagram, YouTube, Discord, ...	Between 2 and 3 hours	5.0	1	1	1	0	1	0	0	1	1	2.5
1	2022-04-18 19:19:20	21.0	Female	Single	University Student	University	True	Facebook, Twitter, Instagram, YouTube, Discord, ...	More than 5 hours	4.0	1	1	1	0	1	0	0	1	1	6.0
2	2022-04-18 19:25:29	21.0	Female	Single	University Student	University	True	Facebook, Instagram, YouTube, Pinterest	Between 3 and 4 hours	3.0	0	1	1	1	0	0	0	0	1	3.5
3	2022-04-18 19:26:42	21.0	Female	Single	University Student	University	True	Facebook, Instagram	More than 5 hours	4.0	0	1	1	0	0	0	0	0	0	6.0
4	2022-04-18 19:33:31	21.0	Female	Single	University Student	University	True	Facebook, Instagram, YouTube	Between 2 and 3 hours	3.0	0	1	1	0	0	0	0	0	1	2.5

Figure 15: First rows of the Social Media and Mental Health dataset (481 rows, 31 columns).

6. Student Social Media & Mental Health

student_sm_clean: (4998, 14)

	age	gender	country	academic_level	most_used_platform	purpose_of_use	avg_daily_usage_hours	daily_unlocks	study_hours	physical_activity_hours	sleep_hours_per_night	stress_level	mental_health_score	stress_level_num
0	21.0	Male	Other	Undergraduate	Facebook	Networking	4.0	134.0	4.5	2.2	6.7	Medium	6.8	2.0
1	23.0	Female	Other	Graduate	Linkedin	Education	1.6	73.0	7.0	2.4	8.6	Low	7.6	1.0
2	22.0	Male	Canada	Undergraduate	Instagram	Entertainment	4.6	166.0	4.0	1.8	6.7	Medium	7.0	2.0
3	18.0	Male	Other	High School	Snapchat	Entertainment	7.0	220.0	1.0	1.7	5.4	Very High	5.3	2.0
4	24.0	Female	Other	Graduate	Facebook	Networking	7.5	237.0	1.0	1.1	5.0	Very High	4.4	2.0

Figure 16: First rows of the Student Social Media & Mental Health dataset (4,998 rows, 14 columns).

5.1.2 Reddit

1. Posts

posts: (48622, 13)

```

subreddit
explainlikeimfive    19289
todayilearned        16153
books                 6223
teenagers             4778
memes                 2121
nosurf                58
Name: count, dtype: int64

```

	created	subreddit	author	title	score	num_comments	domain
683	2016-06-04 10:15:50+00:00	books	trinhlam365	New Hội An city – Tổ hợp du lịch nghỉ dưỡng đắt giá nhất...	1	0	newhoiancity.vietnamcondotel.net
3694	2016-06-18 18:15:38+00:00	books	[deleted]	[META]book nerds on reddit love being validated	1	0	dailydot.com
9178	2016-06-06 00:31:11+00:00	explainlikeimfive	SoleyBored	ELI5: Why do so many people vanish mysteriously in natio...	0	3	self.explainlikeimfive
23203	2016-06-27 02:47:19+00:00	explainlikeimfive	[deleted]	Why can you feel your heart more clearly after physical ...	1	0	self.explainlikeimfive
25828	2016-06-04 17:13:52+00:00	memes	mbmanning	GREATEST OF ALL TIME. #RIPAli	1	0	imgur.com
26510	2016-06-14 09:35:25+00:00	memes	Tyrann0saurus_Rex	About r/the_donald	1	0	imgflip.com
27644	2016-06-05 19:48:52+00:00	nosurf	[deleted]	How many men watch porn?	4	10	self.nosurf
27689	2016-06-28 14:08:14+00:00	nosurf	Kiechi	I've noticed my memory sucks compared to what it used to...	3	3	self.nosurf
28491	2016-06-05 20:03:57+00:00	teenagers	teenagerstrowaway7	[Advice] How do I become more confident again and make b...	7	0	self.teenagers
28595	2016-06-06 04:54:39+00:00	teenagers	128keaton	[Relationship] I asked a girl out and got her number	11	2	self.teenagers
41608	2016-06-17 10:16:31+00:00	todayilearned	Xolotl123	TIL that the BBC regularly rehearses the Queen's death i...	5385	1773	mirror.co.uk
34375	2016-06-04 07:17:12+00:00	todayilearned	pvolk	TIL "Benjamin Bennett" has a YouTube channel with only v...	1	0	youtu.be

Figure 17: Reddit posts from June 2016 (48,622 rows, 13 columns): counts per subreddit and two random posts from each subreddit.

2. Comments

```

comments: (750651, 10)
subreddit
todayilearned      462234
explainlikeimfive  148820
books              71825
teenagers          65492
memes              1989
nosurf             291
Name: count, dtype: int64

```

	created	subreddit	author	score	body	parent_id	link_id
17193	2016-06-08 17:35:14+00:00	books	demopolis	4	Not to mention amazon. used books there are <2\$ ship...	t1_d40xz8d	t3_4n4jnb
53243	2016-06-21 16:20:17+00:00	books	patentolog1st	0	No, they really don't listen to the public.\n\nThey don't...	t1_d4i4cy8	t3_4p1r6h
196490	2016-06-25 15:17:44+00:00	explainlikeimfive	VictorTrejo	3	But it makes sense with context. The previous post was a...	t1_d4njv7m	t3_4poj64
116897	2016-06-10 02:49:00+00:00	explainlikeimfive	slash178	2	Rape is far more common than murder.\n\nMurders of rando...	t3_4ne9yq	t3_4ne9yq
221152	2016-06-07 21:48:20+00:00	memes	Meme_Catalogue	1	Doot doot	t3_4luake	t3_4luake
221442	2016-06-12 14:07:31+00:00	memes	Do_Whatever_You_Like	1	huh?	t1_d45s8zs	t3_4nm91g
222783	2016-06-09 22:08:17+00:00	nosurf	LifeAgain	1	Thanks so much.. my strategy is just go cold turkey on q...	t1_d42py82	t3_4nc4qw
222917	2016-06-28 15:19:04+00:00	nosurf	suzyduzy	5	If by 'cold turkey' you mean cutting yourself off entire...	t3_4q9ik3	t3_4q9ik3
281378	2016-06-28 07:59:41+00:00	teenagers	DullEmber	2	Start a staff revolution, overthrow the main office.	t1_d4qyr2w	t3_4q816d
246738	2016-06-12 15:52:38+00:00	teenagers	owetre18	1	I have friends ranging from 13-25. I'm 16, if you can't ...	t3_4nops8	t3_4nops8
441457	2016-06-10 16:02:47+00:00	todayilearned	HCagn	16	Great episode too. Think it's This American Life - ep 56...	t1_d43okkq	t3_4ng2s1
469395	2016-06-12 14:30:55+00:00	todayilearned	[deleted]	2	To make a spectacle. \n\nSPOILERS AHEAD!\n\nThere is a s...	t1_d45u4g4	t3_4np0ch

Figure 18: Reddit comments from June 2016 (750,651 rows, 10 columns): counts per subreddit and two random comments from each subreddit.

5.2 Metrics

5.2.1 Kaggle

Table 1: Size of the Kaggle datasets after cleaning.

Dataset	File size (MB)	Memory (MB)	Rows	Columns
Doomscrolling_Study2_Dataset	0.11	0.33	401	93
Doomscrolling_Study3_Dataset	0.26	0.84	502	205
reels_attention_span_dataset_12000	1.10	1.63	12,000	11
sleep_doomscrolling_habits_clean	0.17	0.26	1,000	30
smmh_clean	0.10	0.16	481	31
Social_Media_Addiction	0.08	0.15	1,199	11
student_sm_clean	0.42	0.76	4,998	14
Total	2.25	4.12	20,581	395

Frequency:

Table 2: Gender distribution in the Doomscrolling Study 2 (401 rows) and Study 3 (502 rows) datasets.

Column	Value	Study 2		Study 3	
		Count	%	Count	%
Gender	1	242	60.3	292	58.2
	2	157	39.2	208	41.4
	3	2	0.5	2	0.4

Table 3: Category frequencies in the Social Media Addiction dataset (1,199 rows).

Column	Value	Count	%
gender	male	612	51.0
	female	579	48.3
	Unknown	8	0.7
platform_usage	Instagram	411	34.3
	TikTok	397	33.1
	Both	391	32.6
social_interaction_level	low	415	34.6
	medium	415	34.6
	high	369	30.8

Table 4: Category frequencies in the Reels/Shorts Consumption vs Attention Span dataset (12,000 rows).

Column	Value	Count	%
platform	Instagram Reels	4,033	33.6
	TikTok	3,994	33.3
	YouTube Shorts	3,973	33.1
stress_level	High	4,031	33.6
	Medium	3,999	33.3
	Low	3,970	33.1

Table 5: Category frequencies in the Sleep & Doomscrolling Habits dataset (1,000 rows).

Column	Value	Count	%
gender	Male	497	49.7
	Female	450	45.0
	Prefer Not To Say	53	5.3
occupation_status	Employed Full-Time	403	40.3
	Student	346	34.6
	Employed Part-Time	140	14.0
	Unemployed	111	11.1
sleep_quality_category	Good	340	34.0
	Poor	330	33.0
	Fair	330	33.0
doomscroller	False	524	52.4
	True	476	47.6

Table 6: Category frequencies in the Social Media and Mental Health dataset (481 rows).

Column	Value	Count	%
gender	Female	263	54.7
	Male	211	43.9
	Other	7	1.5
occupation_status	University Student	292	60.7
	Salaried Worker	132	27.4
	School Student	49	10.2
	Retired	8	1.7
avg_daily_time_social_media	More than 5 hours	116	24.1
	Between 2 and 3 hours	101	21.0
	Between 3 and 4 hours	93	19.3
	Between 1 and 2 hours	70	14.6
	Between 4 and 5 hours	67	13.9
	Less than an Hour	34	7.1

Table 7: Category frequencies in the Student Social Media & Mental Health dataset (4,998 rows).

Column	Value	Count	%
gender	Male	2,634	52.7
	Female	2,364	47.3
academic_level	Undergraduate	3,630	72.6
	Graduate	918	18.4
	High School	450	9.0
purpose_of_use	Entertainment	2,551	51.0
	Education	1,093	21.9
	Networking	793	15.9
	News	561	11.2
stress_level	Very High	1,620	32.4
	High	1,439	28.8
	Medium	1,295	25.9
	Low	643	12.9
	Ver Highy	1	0.0

5.2.2 Reddit

Extracted from 1.4 TB of compressed Reddit dumps:

Table 8: Total size and number of records of the extracted Reddit data.

Type	Records	Size (MB)
Comments	81,949,500	7,279
Submissions	2,354,131	284
Total	84,303,631	7,563

Table 9: Number of Reddit records per subreddit and type.

Subreddit	Comments	Submissions
r/books	5,525,094	223,754
r/explainlikeimfive	11,278,386	693,580
r/memes	7,184,620	102,389
r/nosurf	40,575	962
r/teenagers	18,458,392	337,015
r/todayilearned	39,462,433	996,431

The numbers above are for the data that has already been downloaded (comments from 2012 to mid-2019, posts from 2012 to 2016).

Estimated numbers for the full dataset:

- Size: 25–30 GB
- Row count: about 350 million

6 Links

- YouTube video
- GitHub
- Datasets

Note: since the Reddit data takes 7.6 GB and will grow to 25–30 GB, it is stored on a separate flash drive.